

部署示例模型

1.部署mlir时无法顺利执行model_transform.py

```
root@9494a6627db5:/workspace/examples/simple/yolact/workspace# sh onnx2mlir.sh
Traceback (most recent call last):
  File "/workspace/tpu-mlir_v1.1.7-g30df2e3c-20230605/python/tools/model_transform.py", line 17, in <module>
    from utils.mlir_parser import *
  File "/workspace/tpu-mlir_v1.1.7-g30df2e3c-20230605/python/utils/mlir_parser.py", line 14, in <module>
    from mlir.ir import *
  File "/workspace/tpu-mlir_v1.1.7-g30df2e3c-20230605/python/mlir/ir.py", line 5, in <module>
    from .mlir_libs._mlir.ir import *
  File "/workspace/tpu-mlir_v1.1.7-g30df2e3c-20230605/python/mlir/_mlir_libs/_init_.py", line 128, in <module>
    _site_initialize()
  File "/workspace/tpu-mlir_v1.1.7-g30df2e3c-20230605/python/mlir/_mlir_libs/_init_.py", line 56, in _site_initialize
    from ._mlir import ir
ImportError: cannot import name 'ir' from 'mlir._mlir_libs._mlir' (unknown location)
```

问题1. 基于YOLACT的目标跟踪算法移植与测试

模型转化阶段：ONNX转MLIR

执行命令：sh onnx2mlir.sh

提示：ImportError:cannotimport name'ir'from 'mlir. mlirlibs.mlir'(unknown location)

尝试解决：问题文件为model_transforme.py，在python中可以import mlir，无法import mlir._mlir_libs，该文件下ir只有ir.pyi。（提示为修改过文件地址之后的，未修改显示no module）

解决：docker的镜像错误

官方部署Llama-tpu

问题2：无法source文件

- 1 | `$'\r': command not found`
- 2 | 原因：windows中的换行符是 \r，linux中的是\n。

- 1 | 通过安装dos2unix解决
- 2 | `sudo apt-get install dos2unix`
- 3 | `dos2unix envsetup.sh`

问题3：./build.sh失败

- 1 | `-bash: /home/xxx.sh: /bin/bash^M: bad interpreter: No such file or directory`

解决：格式问题

- 1 | windows格式的文件行尾为^M\$，unix格式的文件行尾为\$
- 2 | 1.查看文件格式
- 3 | `cat -A xxx.sh`
- 4 | : dos格式的文件行尾为^M\$，unix格式的文件行尾为\$
- 5 | 2.使用下述命令直接替换结尾符为unix格式
- 6 | `sed -i "s/\r//" build.sh`
- 7 | : 再次使用1查看文件行尾

问题4：./build.sh显示权限错误

```
1 | bash: ./build.sh: Permission denied
2
3 | 解决:  chmod 777 build.sh
```

结果

```
step
/bin/sh: 1: /workspace/tpu-mlir/python/utils/merge_pyfbs.py: Permission denied
[48/1233] Building CXX object third_party/cnpy/CMakeFiles/cnpy.dir/cnpy.cpp.o
ninja: build stopped: subcommand failed.
root@805e2fb748f5:/workspace/tpu-mlir#
```

Llama2-TPU

编译模型阶段：导出所有onnx模型

执行命令：python3 export_onnx.py

提示：killed

尝试：执行时修改了源文件的地址，执行时同时查看动态内存，到达内存满的临界状态显示killed。服务器内存15g。

问题： 错误：内存不足

查看sc7使用：bm-smi

*问题5：红色E

需要重新更新

```
1 | sudo apt-get update
```

```
root@4389e82e1c63:/workspace/Release_230701-public/libsonphon_20230806_182258# sudo apt install dkms libncurses5
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package dkms
```

问题：无法加载权重文件

```
root@805e2fb748f5:/workspace/Llama2-TPU/compile# python3 export_onnx.py
Loading checkpoint shards: 50% | 1/2 [00:05<00:05, 5.42s/it]
Traceback (most recent call last):
  File "/usr/local/lib/python3.10/dist-packages/transformers/modeling_utils.py", line 460, in load_state_dict
    return torch.load(checkpoint_file, map_location="cpu")
  File "/usr/local/lib/python3.10/dist-packages/torch/serialization.py", line 797, in load
    with _open_zipfile_reader(opened_file) as opened_zipfile:
  File "/usr/local/lib/python3.10/dist-packages/torch/serialization.py", line 283, in __init__
    super().__init__(torch._C.PyTorchFileReader(name_or_buffer))
RuntimeError: PytorchStreamReader failed reading zip archive: failed finding central directory

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/usr/local/lib/python3.10/dist-packages/transformers/modeling_utils.py", line 464, in load_state_dict
    if f.read(7) == "version":
  File "/usr/lib/python3.10/codecs.py", line 322, in decode
    (result, consumed) = self._buffer_decode(data, self.errors, final)
UnicodeDecodeError: 'utf-8' codec can't decode byte 0x80 in position 128: invalid start byte

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "/workspace/Llama2-TPU/compile/export_onnx.py", line 26, in <module>
    origin_model = LlamaForCausalLM.from_pretrained(model_path)
  File "/usr/local/lib/python3.10/dist-packages/transformers/modeling_utils.py", line 2903, in from_pretrained
    ) = cls._load_pretrained_model(
  File "/usr/local/lib/python3.10/dist-packages/transformers/modeling_utils.py", line 3246, in _load_pretrained_model
    state_dict = load_state_dict(shard_file)
  File "/usr/local/lib/python3.10/dist-packages/transformers/modeling_utils.py", line 476, in load_state_dict
    raise OSError(
OSError: Unable to load weights from pytorch checkpoint file for '../work/llama-2-7b-hf/pytorch_model-00002-of-00002.bin' at '../work/llama-2-7b-hf/pytorch_model-00002-of-00002.bin'. If you tried to load a PyTorch model from a TF 2.0 checkpoint, please set from_tf=True.
```

权重文件下载时出现错误，需要重新下载

问题：/usr/bin/env: 'python3\r': No such file or directory

```
1 | dos2unix model_transform.py
2 | dos2unix model_deploy.py
```

```

root@805e2fb748f5:/workspace/Llama2-TPU/compile# ./compile.sh --num_device 2
+ models=
+ mode=f16
+ num_device=1
+ mode_args=
+ device_args=
+ out_model=llama2-7b.bmodel
+ [[ 2 -gt 0 ]]
+ key=--num_device
+ case $key in
+ num_device=2
+ shift 2
+ [[ 0 -gt 0 ]]
+ '[' x2 '!=' x1 ']'
+ device_args='--num_device 2'
+ out_model=llama2-7b_f16_2dev.bmodel
+ outdir=tmp/embedding
+ mkdir -p tmp/embedding
+ pushd tmp/embedding
/workspace/Llama2-TPU/compile/tmp/embedding /workspace/Llama2-TPU/compile
+ seqlen=512
+ model_transform.py --model_name embedding --model_def ../embedding.onnx --input_shapes '[[512]]' --mlir embedding_512.mlir
/usr/bin/env: 'python3\r': No such file or directory
root@805e2fb748f5:/workspace/Llama2-TPU/compile#

```

问题：无法加载mlir

```

root@805e2fb748f5:/workspace/Llama2-TPU/compile# ./compile.sh --num_device 2
+ models=
+ mode=f16
+ num_device=1
+ mode_args=
+ device_args=
+ out_model=llama2-7b.bmodel
+ [[ 2 -gt 0 ]]
+ key=--num_device
+ case $key in
+ num_device=2
+ shift 2
+ [[ 0 -gt 0 ]]
+ '[' x2 '!=' x1 ']'
+ device_args='--num_device 2'
+ out_model=llama2-7b_f16_2dev.bmodel
+ outdir=tmp/embedding
+ mkdir -p tmp/embedding
+ pushd tmp/embedding
/workspace/Llama2-TPU/compile/tmp/embedding /workspace/Llama2-TPU/compile
+ seqlen=512
+ model_transform.py --model_name embedding --model_def ../embedding.onnx --input_shapes '[[512]]' --mlir embedding_512.mlir
Traceback (most recent call last):
  File "/workspace/tpu-mlir/python/tools/model_transform.py", line 17, in <module>
    from utils.mlir_parser import *
  File "/workspace/tpu-mlir/python/utils/mlir_parser.py", line 12, in <module>
    import mlir
ModuleNotFoundError: No module named 'mlir'
root@805e2fb748f5:/workspace/Llama2-TPU/compile#

```

解决，重复拉去mlir包

问题：error: 'bmrt_create_ex' was not declared in this scope; did you mean 'bmrt_create'?

```

root@cb5a57685a0a:/workspace/Llama2-TPU/demo/build# make -j
[ 50%] Building CXX object CMakeFiles/llama2.dir/demo.cpp.o
/workspace/Llama2-TPU/demo/demo.cpp: In member function 'void Llama2::init(const std::vector<int>&, std::string)':
/workspace/Llama2-TPU/demo/demo.cpp:110:12: error: 'bmrt_create_ex' was not declared in this scope; did you mean 'bmrt_create'?
  110 |     p_bmrt = bmrt_create_ex(handles.data(), device_num);
      |                  ^~~~~~
      |                  bmrt_create
make[2]: *** [CMakeFiles/llama2.dir/build.make:76: CMakeFiles/llama2.dir/demo.cpp.o] Error 1
make[1]: *** [CMakeFiles/Makefile2:83: CMakeFiles/llama2.dir/all] Error 2
make: *** [Makefile:91: all] Error 2
root@cb5a57685a0a:/workspace/Llama2-TPU/demo/build#

```

看来一下api

```

/* ----- */
/**
 * @name    bmrt_create
 * @brief   To create the bmruntime with bm_handle.
 * @ingroup bmruntime
 *
 * This API creates the bmruntime. It returns a void* pointer which is the pointer
 * of bmruntime. Device id is set when get bm_handle;
 *
 * @param [in] bm_handle    bm handle. It must be initialized by using bmlib.
 *
 * @retval void* the pointer of bmruntime
 */
DECL_EXPORT void* bmrt_create(bm_handle_t bm_handle);

```

修改之后

cannot convert 'bm_context*' to 'bm_handle_t' {aka 'bm_context'}

```
root@cb5a57685a0a:/workspace/Llama2-TPU/demo/build# make -j
[ 50%] Building CXX object CMakeFiles/llama2.dir/demo.cpp.o
/workspace/Llama2-TPU/demo/demo.cpp: In member function 'void LLama2::init(const std::vector<int>&, std::string)':
/workspace/Llama2-TPU/demo/demo.cpp:110:36: error: cannot convert 'bm_context**' to 'bm_handle_t' {aka 'bm_context*'}
   110 |     p_bmrt = bmrt_create(handles.data(), device_num);
       |                               ~~~~~~^
                               |
                               bm_context**
In file included from /workspace/Llama2-TPU/demo/demo.cpp:18:
/opt/sophon/libsonphon-current/include/bmruntime_interface.h:88:43: note: initializing argument 1 of 'void* bmrt_create(bm_handle_t
)'
   88 | DECL_EXPORT void* bmrt_create(bm_handle_t bm_handle);
       |                               ~~~~~~^
make[2]: *** [CMakeFiles/llama2.dir/build.make:76: CMakeFiles/llama2.dir/demo.cpp.o] Error 1
make[1]: *** [CMakeFiles/Makefile2:83: CMakeFiles/llama2.dir/all] Error 2
make: *** [Makefile:91: all] Error 2
root@cb5a57685a0a:/workspace/Llama2-TPU/demo/build#
```